Evaluating Uncertainty Estimation: Proper Scoring

Proper Scoring: Definition

- Imagine a scoring function $S(P_{\boldsymbol{\theta}}(y|\mathbf{x}_i), (\mathbf{x}_i, y_i))$ that measures the quality of a predictive distribution $P_{\boldsymbol{\theta}}(y|\mathbf{x})$ for an observed outcome or data $(\mathbf{x}_i, y_i)$ sampled from an underlying true distribution $P(\mathbf{x}, y) = P(y|\mathbf{x})P(\mathbf{x})$.

- Let us define $S(P_{\boldsymbol{\theta}}, P) = \int\int P(\mathbf{x}, y) S(P_{\boldsymbol{\theta}}, (\mathbf{x}, y)) d\mathbf{x} dy$

- S is called a *proper* scoring function if $S(P_{\boldsymbol{\theta}}, P) \leq S(P, P)$.

- *Strictly* proper scoring function with equality *iff* $P_{\boldsymbol{\theta}}(y|\mathbf{x}) = P(y|\mathbf{x})$.

[Brier. Verification of forecasts expressed in terms of probability. Monthly weather review, 1950.]
[Gneiting and Raftery, "Strictly Proper Scoring Rules, Prediction, and Estimation", JASA 2007]

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Proper Scoring: Examples

- Log Likelihood
  - $S(P_{\boldsymbol{\theta}}, (\mathbf{x}_i, y_i)) = log\big(P_{\boldsymbol{\theta}}(y = y_i|\mathbf{x}_i)\big)$
- Brier Score
  - $S(P_{\boldsymbol{\theta}}, (\mathbf{x}_i, y_i)) = MSE\big(P_{\boldsymbol{\theta}}(y|\mathbf{x}_i), \delta_y\big) = \sum_a (P_{\boldsymbol{\theta}}(y = a|\mathbf{x}_i) - \mathbf{1}(a = y_i))^2$

- Other examples
  - Mean Absolute Error (MAE) is improper (it prefers overconfident predictions)
    - $S(P_{\boldsymbol{\theta}}, (\mathbf{x}_i, y_i)) = MAE\big(P_{\boldsymbol{\theta}}(y|\mathbf{x}_i), \delta_y\big) = \sum_a |P_{\boldsymbol{\theta}}(y = a|\mathbf{x}) - \mathbf{1}(a = y_i)|$
  - Accuracy (0-1 loss) is not *strictly* proper
    - $S(P_{\boldsymbol{\theta}}, (\mathbf{x}_i, y_i)) = \mathbf{1}(argmax_a P_{\boldsymbol{\theta}}(y = a|\mathbf{x}) = y_i)$

[Gneiting and Raftery, "Strictly Proper Scoring Rules, Prediction, and Estimation", JASA 2007]



Evaluating Uncertainty Estimation

How to evaluate the quality of uncertainty estimates?

- Proper Scoring Rule (LL, Brier Score)
  - [Gneiting and Raftery, "Strictly Proper Scoring Rules, Prediction, and Estimation", JASA 2007]
- Calibration (NLL, ECE)
  - ["Obtaining Well Calibrated Probabilities Using Bayesian Binning", AAAI 2015]
- Out-of-distribution detection



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Proper Scoring: properties

- Importantly, a proper scoring function encourages
  - high accuracy
  - high calibration

We will define the notion of calibration next

[Brier. Verification of forecasts expressed in terms of probability. Monthly weather review, 1950.]
[Gneiting and Raftery, "Strictly Proper Scoring Rules, Prediction, and Estimation", JASA 2007]